Chapter 16, Problem 95P

Problem

Given the following state equation, solve for $y1(t)$ and $y2(t)$.

$$\dot{\mathbf{x}} = \begin{bmatrix} -2 & -1 \\ 2 & -4 \end{bmatrix} x + \begin{bmatrix} 1 & 1 \\ 4 & 0 \end{bmatrix} \begin{bmatrix} u(t) \\ 2u(t) \end{bmatrix}$$

$$\mathbf{y} = \begin{bmatrix} -2 & -2 \\ 1 & 0 \end{bmatrix} x + \begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} u(t) \\ 2u(t) \end{bmatrix}$$

Step-by-step solution

Step 1 of 2

Assuming the initial conditions are Zero. Using Lap lace transforms we get,

$$X(S) = \begin{bmatrix} S+2 & 1 \\ -2 & S+4 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 1 \\ 4 & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{S} \\ \frac{2}{S} \end{bmatrix} = \frac{1}{S^2+6S+10} \begin{bmatrix} S+4 & -1 \\ 2 & S+2 \end{bmatrix} \begin{bmatrix} \frac{3}{S} \\ \frac{4}{S} \end{bmatrix}$$

$$X_1 = \frac{3S+8}{S\left((S+3)^2+1^2\right)} = \frac{0.8}{S} + \frac{-0.8S-1.8}{(S+3)^2+1^2}$$

$$= \frac{0.8}{S} - \frac{0.8(S+3)}{(S+3)^2+1^2} + \frac{0.6}{(S+3)^2+1^2}$$

Step 2 of 2

$$X_1(t) = \left(0.8 - 0.8e^{-3t}\cos t + 0.6e^{-3t}\sin t\right)u(t)$$

$$X_2 = \frac{4S+14}{S\left((S+3)^2+1^2\right)} = \frac{1.4}{S} + \frac{-1.4S-4.4}{(S+3)^2+1^2}$$

$$= \frac{1.4}{S} - \frac{1.4(S+3)}{(S+3)^2+1^2} - \frac{0.2}{(S+3)^2+1^2}$$

$$X_2(t) = \left(1.4 - 1.4e^{-3t}\cos t - 0.2e^{-3t}\sin t\right)u(t)$$

$$y_1(t) = -2x(t) - 2x_2(t) + 2u(t)$$

$$= \left(-2.4 + 4.4e^{-3t}\cos t - 0.8e^{-3t}\sin t\right)u(t)$$

$$y_2(t) = x_1(t) - 2u(t) = \left(-1.2 - 0.8e^{-3t}\cos t + 0.6e^{-3t}\sin t\right)u(t)$$